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Subject: Calculus II

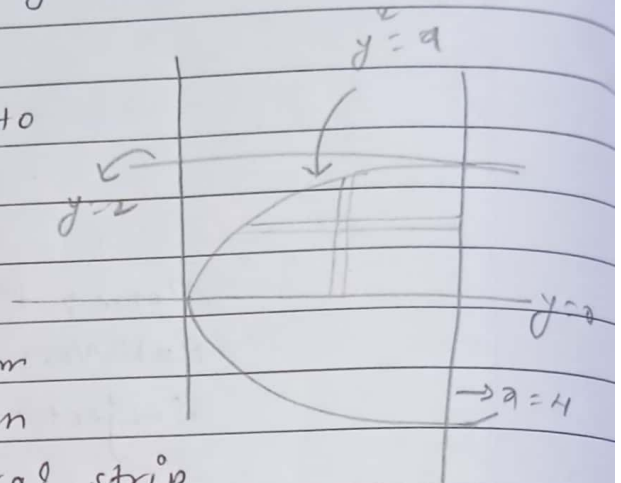
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Model Paper Solution

17 a) Evaluate the integral: $\int_0^2 \int_{y^2}^4 y \cos(x^2) dx dy$

Solⁿ

The curve is bounded by $x = y^2$ to $x = 4$ and $y = 0$ to $y = 2$. Here, the region is first integrated w.r.t. x along horizontal strip which extends from $x = y^2$ to $x = 4$ and then w.r.t. y along vertical strip which extends from $y = 0$ to $y = 2$.



By changing the order of integration, we first integrate w.r.t. y along the vertical strip which extends from $y = 0$ to $y = \sqrt{x}$ and then w.r.t. x along horizontal strip which extends from $x = 0$ to $x = 4$.

Hence,
$$\int_0^4 \int_0^{\sqrt{x}} y \cos x^2 dy dx$$

$$= \int_0^4 \cos x^2 \left[\frac{y^2}{2} \right]_0^{\sqrt{x}} dx$$

$$= \int_0^4 \frac{x}{2} \cos x^2 dx$$

Let, $x^2 = t$

$\therefore 2x dx = dt$

$$x dx = \frac{dt}{2}$$

when $x = 0$ then $t = 0$

$x = 4$ then $t = 16$

So,

$$\begin{aligned}\int_0^4 \frac{t}{2} \cos t^2 dt &= \int_0^{16} \frac{\cos t}{4} dt \\ &= \frac{1}{4} [\sin t]_0^{16} \\ &= \frac{1}{4} \sin 16 \quad \underline{\text{Ans.}}\end{aligned}$$

1 > b >

Solⁿ

Evaluate the integral: $\int_0^1 \int_0^{(1-q)} \int_0^{(q+y)} e^z dz dy dq$.
given,

$$\begin{aligned}&\int_0^1 \int_0^{(1-q)} \int_0^{(q+y)} e^z dz dy dq \\ &= \int_0^1 \int_0^{(1-q)} [e^z]_0^{(q+y)} dy dq \\ &= \int_0^1 \int_0^{(1-q)} (e^{(q+y)} - 1) dy dq \\ &= \int_0^1 e^q [e^y - y]_0^{1-q} dq \\ &= \int_0^1 \{e^q [e^{1-q} - 1] - 1 + q\} dq \\ &= \int_0^1 (e - e^q - 1 + q) dq \\ &= \left[e q - e^q - q + \frac{q^2}{2} \right]_0^1\end{aligned}$$

$$= (e - e - 1 + \frac{1}{2} + 1)$$

$$= \frac{1}{2} \text{ Ans.}$$

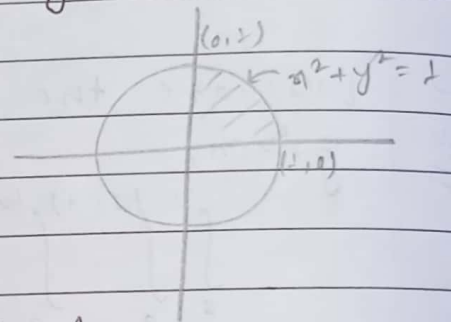
1>c> Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2 + y^2 = 1$ and the plane $z + y = 3$.

Soln

Here, plane is,

$$z + y = 3$$

$$\therefore z = 3 - y$$

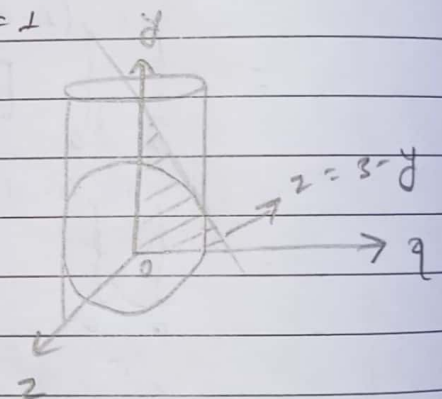


We have integrate $z = 3 - y$ over the first quadrant base of cylinder of the circle $x^2 + y^2 = 1$

For the base, limits for y ,

$$y = 0 \text{ to } y = \sqrt{1 - x^2}$$

and range of x is $0 \leq x \leq 1$ then,



$$V = \int_0^1 \int_0^{\sqrt{1-x^2}} z \, dy \, dx \quad \text{--- (i)}$$

For polar form, let,

$$y = r \sin \theta, \quad x = r \cos \theta$$

$$dy \, dx = r \, dr \, d\theta$$

$$z = 3 - y = 3 - r \sin \theta$$

In first octant, range of θ is,

$$0 \leq \theta \leq \pi/2$$

range of r $0 \leq r \leq 1$

then eqⁿ (i) becomes,

$$V = \int_0^{\pi/2} \int_0^1 (3 - \sin \theta) r \, dr \, d\theta$$

$$= \int_0^{\pi/2} \left[\frac{3r^2}{2} - \sin \theta \frac{r^2}{2} \right]_0^1 d\theta$$

$$= \int_0^{\pi/2} \left[\frac{3}{2} - \frac{\sin \theta}{2} \right] d\theta$$

$$= \left[\frac{3\theta}{2} + \frac{\cos \theta}{2} \right]_0^{\pi/2}$$

$$= \frac{3}{2} \cdot \frac{\pi}{2} + \frac{1}{2} (\cos \frac{\pi}{2} - \cos 0)$$

$$= \frac{3\pi}{4} - \frac{1}{2} \text{ Ans.}$$

Q.2 > Solve by using power series: $y'' - 4xy' + (4x^2 - 2)y = 0$
Soln given,

The differential eqn is,

$$y'' - 4xy' + (4x^2 - 2)y = 0 \quad \text{--- (i)}$$

$$\text{Let, } y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots \quad \text{--- (ii)}$$

be the solution of eqn (i)

$$\text{then, } y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \dots$$

$$\text{Also, } y'' = 2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 \quad \text{--- (iii)}$$

--- (iv)

Now Replacing the values of (iii), (iii) & (iv) in eqn (i) we get,

$$(2a_2 + 6a_3x + 12a_4x^2 + \dots) - 4x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots) + (4x^2 - 2)(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots) = 0$$

$$\Rightarrow (2a_2 - 2a_0) + (6a_3 - 4a_1 - 2a_2)x + (12a_4 - 8a_2 + 4a_0 - 2a_2)x^2 + (20a_5 - 12a_3 + 4a_1 - 2a_3)x^3 = 0$$

Now, comparing the coefficient of like terms,

$$\begin{aligned} 2a_2 - 2a_0 &= 0 & 6a_3 - 4a_1 - 2a_2 &= 0 & 12a_4 - 8a_2 + 4a_0 - 2a_2 &= 0 \\ [\therefore a_2 = a_0] & & [a_3 = a_1] & & \Rightarrow 12a_4 - 6a_0 &= 0 \\ & & & & [\therefore a_4 = \frac{1}{2}a_0] \end{aligned}$$

$$20a_5 - 12a_3 + 4a_1 - 2a_3 = 0$$

$$\Rightarrow 20a_5 = 14a_1 - 4a_1$$

$$[\therefore a_5 = \frac{1}{2}a_1]$$

Now, Replacing the values of $a_0, a_1, a_2, a_3, a_4, a_5$,

$$y = a_0 + a_1x + a_0x^2 + a_1x^3 + \frac{1}{2}a_0x^4 + \frac{1}{2}a_1x^5 + \dots$$

$$\Rightarrow y = a_0 \left(1 + x^2 + \frac{1}{2}x^4 + \dots \right) + a_1 \left(1 + x^3 + \frac{1}{2}x^5 + \dots \right)$$

which is the required soln.

Q.7 b) i) Express, $2x^2 - 4x + 2$ as Legendre polynomial

Soln

Here,

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{3x^2 - 1}{2}$$

$$\Rightarrow x^2 = \frac{2P_2(x) + 1}{3}$$

$$= \frac{2P_2(x) + P_0(x)}{3}$$

$$\begin{aligned}\therefore 2q^2 - 4q + 2 &= 2 \left(\frac{2P_2(a) + P_0(a)}{3} \right) - 4P_2(a) + 2P_0(a) \\ &= \frac{4}{3}P_2(a) + \frac{2}{3}P_0(a) - 4P_2(a) + 2P_0(a) \\ &= \frac{4}{3}P_2(a) - 4P_2(a) + \frac{8}{3}P_0(a) \\ &= \frac{4}{3} [P_2(a) - 3P_2(a) + 2P_0(a)]\end{aligned}$$

Q.7 > i) Show that: $J_{\frac{1}{2}}(a) = \sqrt{\frac{2}{\pi a}} \sin a$

Solⁿ Here, the Bessel function is,

$$J_\nu(a) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(\nu+m+1)} \left(\frac{a}{2}\right)^{\nu+2m} \quad \dots (1)$$

Taking $\nu = \frac{1}{2}$, we have,

$$J_{\frac{1}{2}}(a) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\frac{1}{2}+1)} \left(\frac{a}{2}\right)^{2m+\frac{1}{2}}$$

$$= \sqrt{\frac{2}{a}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\frac{3}{2})} \left(\frac{a}{2}\right)^{2m+1}$$

$$\therefore J_{\frac{1}{2}}(a) = \sqrt{\frac{2}{a}} \sum_{m=0}^{\infty} \frac{(-1)^m}{2^{2m+1} m! \Gamma(m+\frac{3}{2})} a^{2m+1} \quad \dots (4)$$

Now, $2^{2m} m! = 2m(2m-2)(2m-4) \dots 4 \cdot 2$

and $2^{2m+1} \Gamma(m+\frac{3}{2}) = 2^{m+1} (m+\frac{1}{2})(m-\frac{1}{2}) \dots \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}$

$$= (2m+1)(2m-1) \cdot 3 \cdot 1 \cdot \sqrt{\pi}$$

from (4)

$$J_{\frac{1}{2}}(a) = \sqrt{\frac{2}{\pi a}} \sum_{m=0}^{\infty} \frac{(-1)^m a^{2m+1}}{(2m+1)!} = \sqrt{\frac{2}{\pi a}} \sin a$$

$$\therefore J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x \quad \#$$

3> a> i> State first shifting theorem of Laplace Transform and find the Laplace transform of $t \cos at$.

Ans:

If the Laplace transform of $f(t)$ is $F(s)$, then Laplace transform of $[e^{at} f(t)]$ is $F(s-a)$ i.e.

if $\mathcal{L}\{f(t)\} = F(s)$ then

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a).$$

This is the 1st shifting theorem of Laplace.

Solution: given,

$t \cos at$

$$\mathcal{L}\{t \cos at\} = -\frac{d}{ds} \mathcal{L}\{\cos at\}$$

$$= -\frac{d}{ds} \left(\frac{s}{s^2 + a^2} \right)$$

$$= \frac{s^2 - a^2}{(s^2 + a^2)^2}$$

3> a>

ii> Find the inverse Laplace transform of the function

$$\frac{s+2}{s^2(s+3)}$$

solⁿ Here,

$$F(s) = \frac{s+1}{s^2(s+3)}$$

let,

$$F(s) = \frac{s+1}{s^2(s+3)} = \frac{As+B}{s^2} + \frac{C}{s+3}$$

then,

$$(As+B)(s+3) + Cs^2 = s+1$$

$$\Rightarrow As^2 + 3As + Cs^2 + Bs + 3B = s+1$$

$$\Rightarrow (A+C)s^2 + (3A+B)s + 3B = s+1$$

Equating the like terms,

$$A+C=0, \quad 3A+B=1, \quad 3B=1$$

$$\Rightarrow \frac{2}{9} + C = 0 \quad \Rightarrow 3A = 1 - \frac{1}{3} \quad \left[\because B = \frac{1}{3} \right]$$

$$\left[\because C = -\frac{2}{9} \right] \quad \left[\because A = \frac{2}{9} \right]$$

$$\therefore F(s) = \frac{\frac{2}{9}s + \frac{1}{3}}{s^2} - \frac{\frac{2}{9}}{s+3}$$

$$= \frac{2s+3}{9s^2} - \frac{2}{9(s+3)}$$

$$= \frac{2s}{9s^2} + \frac{3}{9s^2} - \frac{2}{9(s+3)}$$

$$= \frac{2}{9} \cdot \frac{1}{s} + \frac{3}{9} \cdot \frac{1}{s^2} - \frac{2}{9} \cdot \frac{1}{(s+3)}$$

taking inverse laplace,

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2}{9}\right\}\left(\frac{1}{s}\right) + \mathcal{L}^{-1}\left\{\frac{3}{9} \cdot \frac{1}{s^2}\right\} - \frac{2}{9} \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$

$$= \frac{2}{9} \cdot 1 + \frac{1}{3} t - \frac{2}{9} e^{-3t}$$

$$\therefore F(s) = \frac{2}{9} + \frac{1}{3} t - \frac{2}{9} e^{-3t} \quad \underline{\text{Ans}}$$

3>b> Apply laplace transform to solve the initial value problem : $y'' + 4y' + 3y = e^{-t}$, $y(0) = y'(0) = 1$.

Solⁿ

Here,

$$y'' + 4y' + 3y = e^{-t}$$

Taking laplace transform,

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = \mathcal{L}\{e^{-t}\}$$

$$\Rightarrow s^2 \mathcal{L}\{y\} - sy(0) - y'(0) + 4[s\mathcal{L}\{y\} - y(0)] + 3\mathcal{L}\{y\} = \frac{1}{s+1}$$

$$\Rightarrow \mathcal{L}\{y\} [s^2 + 4s + 3] - s(1) - 1 - 4(1) = \frac{1}{s+1}$$

$$\Rightarrow \mathcal{L}\{y\} = \frac{1 + (s+5)/(s+1)}{(s+2)(s^2 + 4s + 3)}$$

$$= \frac{1 + s^2 + s + s^3 + s^4}{(s+2)(s^2 + 3s + s + 3)}$$

$$= \frac{s^2 + 6s + 6}{(s+2)(s+1)(s+3)}$$

$$= \frac{s^2 + 6s + 6}{(s+2)^2 (s+3)}$$

Using partial fraction,

$$\frac{A}{(s+3)} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2} = \frac{s^2 + 6s + 6}{(s+3)(s+1)^2}$$

$$\Rightarrow A(s+1)^2 + B(s+3)(s+1) + C(s+3) = s^2 + 6s + 6$$

Equating,

$$A+B=1, \quad 2A+4B+C=6, \quad A+3B+3C=6$$

Solving,

$$A = -3/4, \quad B = 7/4, \quad C = 1/2$$

Putting these values,

$$-\frac{3}{4(s+3)} + \frac{7}{4(s+1)} + \frac{1}{2(s+1)^2} = \mathcal{L}\{y\}$$

$$\Rightarrow \mathcal{L}\{y\} = -\frac{3}{4} \mathcal{L}\left\{\frac{1}{s+3}\right\} + \frac{7}{4} \mathcal{L}\left\{\frac{1}{s+1}\right\} + \frac{1}{2} \mathcal{L}\left\{\frac{1}{(s+1)^2}\right\}$$

$$\Rightarrow y = -\frac{3}{4} \mathcal{L}\left\{\frac{1}{s+3}\right\} + \frac{7}{4} \mathcal{L}\left\{\frac{1}{s+1}\right\} + \frac{1}{2} \mathcal{L}\left\{\frac{1}{(s+1)^2}\right\}$$

$$= -\frac{3}{4} \cdot e^{-3t} + \frac{7}{4} e^{-t} + \frac{1}{2} t \cdot e^{-t}$$

$$\therefore y = \frac{7e^{-t} - 3e^{-3t} + 2te^{-t}}{4} \quad \#$$

4>a> A particle moves along the curve $(t^3+1, t^2, 2t+5)$. Find the component of the velocity and acceleration at $t=1$ along $\vec{i} + \vec{j} + 3\vec{k}$.

Solⁿ

Here,

$$\vec{r} = (t^3+1, t^2, 2t+5)$$

$$\vec{r} = (t^3+1)\vec{i} + t^2\vec{j} + (2t+5)\vec{k} \quad \text{--- (i)}$$

Diff. w.r. to t ,

$$\text{Velocity } (\vec{v}) = \frac{d\vec{r}}{dt} = 3t^2\vec{i} + 2t\vec{j} + 2\vec{k} \quad \text{--- (ii)}$$

again, diff. w.r. to t ,

$$\text{acceleration } (\vec{a}) = \frac{d^2\vec{r}}{dt^2} = 6t\vec{i} + 2\vec{j} \quad \text{--- (iii)}$$

given,

$$\vec{n} = \vec{i} + \vec{j} + 3\vec{k}$$

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{\vec{i} + \vec{j} + 3\vec{k}}{\sqrt{1+1+9}} = \frac{1}{\sqrt{11}} (\vec{i} + \vec{j} + 3\vec{k})$$

Component of velocity at $t=1$ then,

$$(3t^2\vec{i} + 2t\vec{j} + 2\vec{k}) \cdot \frac{\vec{i} + \vec{j} + 3\vec{k}}{\sqrt{11}}$$

$$= \frac{3+2+6}{\sqrt{11}} = \sqrt{11}$$

Component of acceleration at $t=1$ then,

$$(6t\vec{i} + 2\vec{j}) \cdot \frac{(\vec{i} + \vec{j} + 3\vec{k})}{\sqrt{11}}$$

$$= \frac{6+2}{\sqrt{11}} = \frac{8}{\sqrt{11}} \quad 4$$

4>b> If $\phi = \ln(x^2 + y^2 + z^2)$, then, find grad ϕ and div (grad ϕ).

Solⁿ Here,

$$\phi = \ln(x^2 + y^2 + z^2)$$

We have to find,

$$\text{grad } \phi = ?$$

$$\text{div}(\text{grad } \phi) = ?$$

Now

$$\begin{aligned} \text{grad } \phi &= \nabla \cdot \phi = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (\ln(x^2 + y^2 + z^2)) \\ &= \frac{2x}{x^2 + y^2 + z^2} \vec{i} + \frac{2y}{x^2 + y^2 + z^2} \vec{j} + \frac{2z}{x^2 + y^2 + z^2} \vec{k} \end{aligned}$$

Again,

$$\text{div}(\text{grad } \phi) = \nabla \cdot \text{grad } \phi$$

$$\begin{aligned} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \left\{ \frac{2x}{x^2 + y^2 + z^2} \vec{i} \right. \\ &\quad \left. + \frac{2y}{x^2 + y^2 + z^2} \vec{j} + \frac{2z}{x^2 + y^2 + z^2} \vec{k} \right\} \end{aligned}$$

$$\begin{aligned} &= \frac{(x^2 + y^2 + z^2) \cdot 2 - 2x \cdot 2x}{(x^2 + y^2 + z^2)^2} + \frac{(x^2 + y^2 + z^2) \cdot 2 - 2y \cdot 2y}{(x^2 + y^2 + z^2)^2} \\ &\quad + \frac{(x^2 + y^2 + z^2) \cdot 2 - 2z \cdot 2z}{(x^2 + y^2 + z^2)^2} \end{aligned}$$

$$= \frac{2(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^2}$$

$$\therefore \operatorname{div}(\operatorname{grad} \phi) = \frac{2}{(x^2 + y^2 + z^2)} \quad \#$$

4>c) Evaluate $\oint_c (x^2 - 3y) dx + (x + \sin y) dy$ where

c is the boundary of the triangle with vertices $(0,0)$, $(1,0)$ and $(0,2)$.

Solⁿ

Here,

$$I = \oint_c (x^2 - 3y) dx + (x + \sin y) dy \quad \text{where } c$$

is the boundary of the triangle.

Here,

$$I = \oint_c (x^2 - 3y) dx + (x + \sin y) dy \quad \text{--- (i)}$$

Comparing (i) with $\oint F_1 dx + F_2 dy$
where,

$$F_1 = (x^2 - 3y) \quad F_2 = (x + \sin y)$$

$$\frac{\partial F_1}{\partial y} = -3$$

$$\frac{\partial F_2}{\partial x} = 1$$

from Green's theorem,

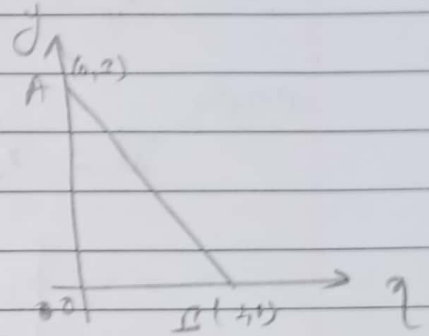
$$\oint F_1 dx + F_2 dy = \iint_S \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA$$

$$= \iint (u+z) \, du \, dy$$

$$= \iint u \, du \, dy$$

Eqⁿ of line AB,
 $y = -2x + 2$

$$\therefore 2x + y = 2$$



Hence, the limit of y is 0 to $2-2x$
the limit of x is 0 to 1.

So,

$$\int_0^1 \int_0^{2-2x} u \, dy \, dx$$

$$= u \int_0^1 (2-2x) \, dx$$

$$= u \left[2x - x^2 \right]_0^1$$

$$= u (2 - 1) = u$$

5>a> find $\iint_S (\vec{F} \cdot \vec{n}) \, dS$, for $\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$,

$$\vec{r} = (u \cos v, u \sin v, uv); 0 \leq u \leq 1, 0 \leq v \leq 2\pi$$

Solⁿ

Here,

$$\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$$

$$\vec{r} = (u \cos v, u \sin v, v)$$

Now,

$$\vec{r}_u = \cos v \vec{i} + \sin v \vec{j}$$

$$\vec{r}_v = -u \sin v \vec{i} + u \cos v \vec{j} + \vec{k}$$

and,

$$\vec{N} = \vec{r}_u \times \vec{r}_v$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos v & \sin v & 0 \\ -u \sin v & u \cos v & 1 \end{vmatrix}$$

$$= 3 \sin v \vec{i} - 3 \cos v \vec{j} + \vec{k} u$$

Now,

$$\vec{F} = x^2 \vec{i} + y^2 \vec{j} + z^2 \vec{k}$$

$$= u^2 \cos^2 v \vec{i} + u^2 \sin^2 v \vec{j} + v^2 \vec{k}$$

$$\vec{F} \cdot \vec{N} = 3u^2 \sin v \cos^2 v - 3u^2 \cos v \sin^2 v + uv^2$$

So,

$$\iint (\vec{F} \cdot \vec{n}) dS = \iint \vec{F} \cdot \vec{N} du dv$$

$$= \int_0^{2\pi} \int_0^1 (3u^2 \sin v \cos^2 v - 3u^2 \cos v \sin^2 v + uv^2) du dv$$

$$= \int_0^{2\pi} \left(u^3 \sin v \cos^2 v - \cos v \sin^2 v u^3 + \frac{uv^2 u^2}{2} \right) dv$$

$$= \int_0^{2\pi} \left(\sin v \cos^2 v - \cos v \sin^2 v + \frac{v^2}{2} \right) dv$$

$$= \left(-\frac{\cos^3 v}{3} \right)_0^{2\pi} - \left(\frac{\sin^3 v}{3} \right)_0^{2\pi} + \frac{2}{2} \left[\frac{v^3}{3} \right]_0^{2\pi}$$

$$= 22\pi^3 \quad \#$$

50b) Evaluate $\oint_C (\vec{F} \cdot d\vec{r})$ by using stoke's theorem,

where $\vec{F} = (y, xz^3, -2yz^3)$ & $C: x^2 + y^2 = 4, z = 3$

Solⁿ Here,

$$\vec{F} = y\vec{i} + xz^3\vec{j} - 2yz^3\vec{k}$$

Now,

$$\text{curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & xz^3 & -2yz^3 \end{vmatrix}$$

$$= \vec{i}(-9y^2 - 0) - 0 + \vec{k}(2x - 1)$$

$$= -9y^2\vec{i} + 26\vec{k}$$

let,

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

also,

$$\vec{r}_x = \vec{i}$$

$$\vec{r}_y = \vec{j}$$

$$\vec{N} = \vec{r}_x \times \vec{r}_y = \vec{k}$$

$$\therefore \text{curl } \vec{F} \cdot \vec{N} = 26$$

Now,

$$\oint_C \vec{F} \cdot d\vec{r} = \iiint \text{curl}(\vec{F} \cdot \vec{N}) \, dx \, dy$$

$$= \iiint 26 \, dx \, dy$$

$$= 26 \cdot \text{area of circle}$$

$$= 26 \cdot (2)^2 \pi$$

$$= 104\pi \quad \#$$

Q.7 > Find the Fourier series of $f(x) = \frac{x^2}{2}$ for $-\pi \leq x \leq \pi$ and deduce that $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots = \frac{\pi^2}{6}$.

Solⁿ

Here,

$$f(x) = \frac{x^2}{2} \quad \text{for } -\pi \leq x \leq \pi$$

Fourier series is,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{--- (1)}$$

where,

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{x^2}{2} \, dx.$$

$$= \frac{1}{2\pi} \cdot \frac{1}{2} \cdot \frac{1}{3} \left[x^3 \right]_{-\pi}^{\pi}$$

$$= \frac{\pi^2}{6}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx,$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{x^2}{2} \cos nx \, dx$$

$$= \frac{1}{2\pi} \cdot 2 \int_0^{\pi} x^2 \cos nx \, dx$$

$$= \frac{1}{\pi} \left[x^2 \frac{\sin nx}{n} + 2x \frac{\cos nx}{n^2} + 2 \frac{\sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\left(0 + \frac{2\pi \cos n\pi}{n^2} - 0 \right) - (0) \right]$$

$$= \frac{2 \cos n\pi}{n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

$$= 0$$

Now eqⁿ (i) becomes,

$$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \left(\frac{2 \cos n\pi}{n^2} \cos nx \right)$$

$$\Rightarrow \frac{q^2}{2} = \frac{\pi^2}{6} + 2 \sum_{n=1}^{\infty} \left(\frac{\cos n\pi \cos nq}{n^2} \right)$$

put $q = \pi$

$$\frac{\pi^2}{2} - \frac{\pi^2}{6} = 2 \left[\frac{\cos \pi \cos \pi}{1^2} + \frac{\cos^2 \pi \cos^2 \pi}{2^2} + \dots \right]$$

$$\Rightarrow \frac{2\pi^2}{6} = 2 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\therefore \frac{\pi^2}{6} = \frac{1}{1} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \quad \#$$

6>b) Find the Fourier half range cosine and sine series of $f(x) = e^x$ in $(0, L)$

Solⁿ

Here,

$$f(x) = e^x$$

Half Range cosine,

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} \right) \quad \text{--- (i)}$$

where,

$$a_0 = \frac{1}{L} \int_0^L e^x dx.$$

$$= \frac{1}{L} \left[e^x \right]_0^L$$

$$= \frac{1}{L} (e^L - 1)$$

$$a_n = \frac{2}{L} \int_0^L e^{ax} \cos\left(\frac{n\pi x}{L}\right) dx.$$

$$= \frac{2}{L} \left[\frac{e^{ax}}{a^2 + \left(\frac{n\pi}{L}\right)^2} \left\{ a \cdot \cos\left(\frac{n\pi x}{L}\right) + \frac{n\pi}{L} \sin\left(\frac{n\pi x}{L}\right) \right\} \right]_0^L$$

$$= \frac{2}{L} \left(\frac{L^2}{L^2 + n^2 \pi^2} \right) \left[e^L \left(\cos n\pi + \frac{n\pi}{L} \sin n\pi \right) - e^0 \left(\cos 0 + \frac{n\pi}{L} \sin 0 \right) \right]$$

$$= \frac{2L}{L^2 + n^2 \pi^2} [e^L (-1)^n + 0 - 1]$$

$$= \frac{2L}{L^2 + n^2 \pi^2} (e^L (-1)^n - 1)$$

eqn (e) becomes,

$$e^x = \frac{1}{L} (e^L - 1) + \sum_{n=1}^{\infty} \left(\frac{2L}{L^2 + n^2 \pi^2} \right) [e^L (-1)^n - 1] \cos\left(\frac{n\pi x}{L}\right).$$

Half Range sine,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) - \left(\frac{\infty}{11}\right)$$

where,

$$b_n = \frac{2}{L} \int_0^L e^{ax} \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{L} \left[\frac{e^{ax}}{a^2 + \left(\frac{n\pi}{L}\right)^2} \left(a \cdot \sin\left(\frac{n\pi x}{L}\right) - \frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) \right) \right]_0^L$$

$$= \frac{d}{L} \left(\frac{L^2}{L^2 + n^2 \pi^2} \right) \left[e^z (\sin n\pi - \frac{n\pi}{L} \cos n\pi) \right] - e^0 (\sin 0 - \frac{n\pi}{L} \cos 0)$$

$$= \frac{2L}{L^2 + n^2 \pi^2} \left(\frac{n\pi}{L} (-1)^n e^L + 1 \right)$$

$$= \frac{2n\pi}{L^2 + n^2 \pi^2} \left[1 - (-1)^n e^L \right]$$

eqⁿ (ii) becomes,

$$e^{\eta} = \sum_{n=1}^{\infty} \frac{2n\pi}{L^2 + n^2 \pi^2} \left[1 - e^L (-1)^n \right] \sin \left(\frac{n\pi \eta}{L} \right)$$

7> Attempt any two questions.

b> Find the breaking time for $u_t + 2u \cdot u_x = 0$,
 $u(x, 0) = e^{-x^2}$

Solⁿ

Here,

$$f(\lambda) = c(u_0(\lambda)) = 2e^{-\lambda^2}$$

$$\therefore f'(\lambda) = \frac{d}{d\lambda} 2e^{-\lambda^2}$$

$$= -4\lambda e^{-\lambda^2}$$

$$t = \frac{-1}{f'(\lambda)} = \frac{1}{-4\lambda e^{-\lambda^2}} = \frac{e^{\lambda^2}}{4\lambda}$$

We need to find the max. value of $t = \frac{e^{\lambda^2}}{4\lambda}$

To find the min. set $\frac{\partial t}{\partial \lambda} = 0$

$$\therefore \frac{4\lambda \cdot e^{\lambda^2} \cdot 2\lambda - e^{\lambda^2} \cdot 4}{16\lambda^2} = 0$$

$$e^{\lambda^2} 4(2\lambda^2 - 1) = 0$$

But,

$4e^{\lambda^2} \neq 0$, we have,

$$2\lambda^2 - 1 = 0$$

$$\lambda^2 = \frac{1}{2}$$

$$\therefore \lambda = \frac{1}{\sqrt{2}}$$

Here, $\frac{d^2 t}{d\lambda^2} > 0$ at $\lambda = \frac{1}{\sqrt{2}}$. So, min. occurs at

$$\lambda = \frac{1}{\sqrt{2}}$$

Hence, breaking time is $t_b = t(\lambda = \frac{1}{\sqrt{2}})$

$$= \frac{e^{\frac{1}{2}}}{4 \cdot \sqrt{\frac{1}{2}}} = \frac{\sqrt{e}}{2\sqrt{2}}$$

7>Q> Evaluate $\int_{(1,0,2)}^{(-2,1,3)} [8xy^3 + 2z^2] dx + 3x^2y^2 dy + (4xz+1) dz$

Solⁿ

Here,

$$I = \int_{(1,0,2)}^{(-2,1,3)} [8xy^3 + 2z^2] dx + 3x^2y^2 dy + (4xz+1) dz$$

where,

$$f_1 = 8xy^3 + 2z^2$$

$$f_2 = 3x^2y^2$$

$$f_3 = 4xz+1$$

Here,

$$\frac{\partial f_1}{\partial y} = 24xz, \quad \& \quad \frac{\partial f_2}{\partial z} = 4x^2y^2 \Rightarrow \frac{\partial f_1}{\partial y} = \frac{\partial f_2}{\partial z}$$

$$\frac{\partial f_1}{\partial z} = 4x \quad \& \quad \frac{\partial f_3}{\partial y} = 4x \Rightarrow \frac{\partial f_1}{\partial z} = \frac{\partial f_3}{\partial y}$$

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$$\frac{\partial f_2}{\partial z} = 0 \text{ \& \ } \frac{\partial f_3}{\partial y} = 0 \Rightarrow \frac{\partial f_2}{\partial z} = \frac{\partial f_3}{\partial y}$$

So, given integral is exact and solution is,

$$I = \int_{(1,0,2)}^{(-2,1,8)} d[f_1 + \int (\text{term free from } x \text{ in } f_2) dy + \int (\text{term free from } x \& y \text{ in } f_3) dz]$$

$$= \int_{(1,0,2)}^{(-2,1,8)} d[\int (6xy^2 + 2z^2) dx + 0 + \int 1 dz]$$

$$= \left[6 \cdot \frac{x^2}{2} y^2 + 2z^2 x + z \right]_{(1,0,2)}^{(-2,1,8)}$$

$$= (12 - 36 + 3) - (8 + 2)$$

$$= -31 \text{ \underline{Ans.}}$$